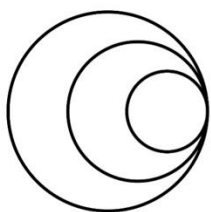




Multilocus sequence typing (MLST) and cgMLST prediction from bacterial genomes

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Multi-locus Sequence Typing (MLST)

- MLST is a method that assigns bacterial isolates to **sequence types (STs)** based on the **allelic profile** of typically **seven** housekeeping genes
 - e.g. *rpoB*, *gapA*, *mdh*, *pgi*, *phoE*, *infB*, *tonB* for *Klebsiella*

Key features of MLST:

- Uses internal fragments of conserved “housekeeping” genes.
- Each unique allele at each locus is assigned a number.
- The combination of allele numbers across all loci defines the ST (e.g. ST78 for *Klebsiella pneumoniae*).
- Databases hosted via **BIGSdb** at **pubmlst.org**, incorporated into many tools
- Highly portable, reproducible, standardized — data comparable across labs and databases.

Multi-locus Sequence Typing (MLST)

MLST database made up of:

- i. set of all allele sequences
- ii. lookup table of allele number combinations to ST

ST	<i>gapA</i>	<i>infB</i>	<i>mdh</i>	<i>pgi</i>	<i>phoE</i>	<i>rpoB</i>	<i>tonB</i>
<u>1</u>	4	4	1	1	7	4	10
<u>2</u>	3	4	1	1	9	4	17
<u>3</u>	5	5	1	1	9	6	11
<u>4</u>	3	1	1	1	3	3	1
<u>5</u>	2	2	1	1	3	3	3

***Klebsiella* MLST scheme**

- Currently has >7500 unique allelic profiles

Strengths & Limitations of MLST

- **Strengths:**

- Standardized, portable typing.
- Good for population structure, global epidemiology, lineage assignment.

- **Limitations:**

- Only a small fraction of the genome (~7 loci) is interrogated → limited resolution.
- Not ideal for distinguishing very closely related isolates (e.g. in outbreak investigations).

Gene-by-Gene approaches

- With whole-genome sequencing (WGS), we no longer need to limit to a few housekeeping genes.
- **Gene-by-gene (GbG)** typing compares genomes using a predefined schema of loci (genes) and assigns alleles per locus, analogous to MLST but on a larger scale.
- Schemas can be:
 - **Core genome** (genes present in nearly all isolates) → cgMLST
 - **Pan-genome / whole genome** → wgMLST or cgMLST/accessory-genome MLST

Gene-by-Gene approaches - cgMLST

Core-genome MLST (cgMLST) expands the MLST concept to hundreds to thousands of loci shared across most or all genomes of a species

Benefits:

- Much greater resolution – reveals fine-scale relationships between isolates.
- Still standardized and portable (allele-based) across labs
- Useful for outbreak detection, transmission mapping, and detailed epidemiology

cgMLST using chewBBACA

- chewBBACA provides schema creation and allele calling for genome-by-genome typing
- Works on draft or complete genomes (assemblies) — suitable for labs without reference-based mapping
- Enables reproducible, scalable cgMLST workflows even on modest computing resources

MICROBIAL GENOMICS

METHODS PAPER

Silva *et al.*, *Microbial Genomics* 2018;4

DOI 10.1099/mgen.0.000166



chewBBACA: A complete suite for gene-by-gene schema creation and strain identification

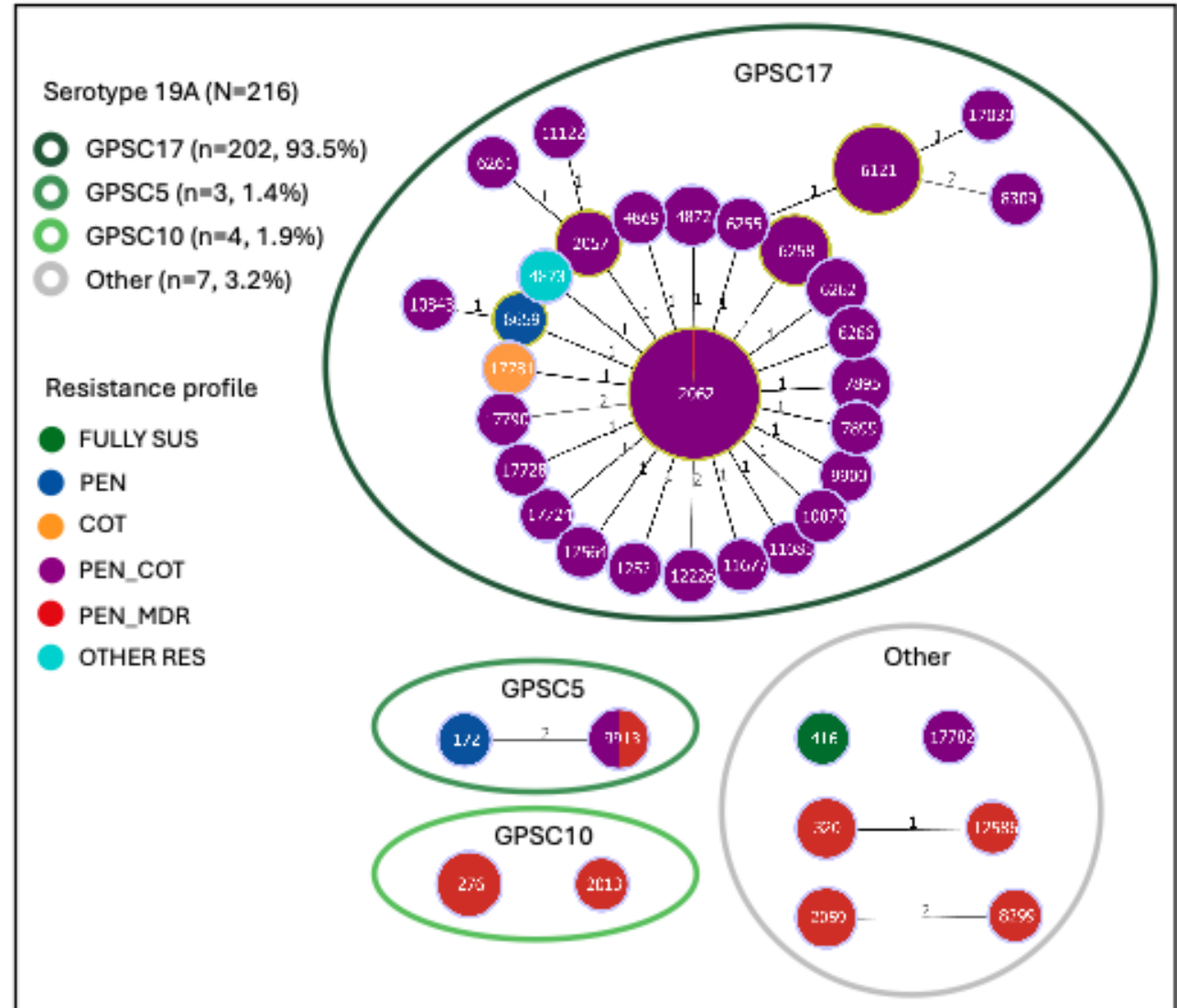
Mickael Silva,¹ Miguel P. Machado,¹ Diogo N. Silva,¹ Mirko Rossi,² Jacob Moran-Gilad,^{3,4} Sergio Santos,¹ Mario Ramirez¹ and João André Carriço^{1,*}

When to Use MLST vs cgMLST

Scenario	Preferred method
Global population structure, long-term epidemiology, community-level surveillance	MLST
High-resolution outbreak investigation, transmission mapping, hospital surveillance	cgMLST
Resource-limited settings with limited genome data	MLST or small-scheme cgMLST
Comparative genomics across diverse strains / broad sampling	cgMLST (or wgMLST)

MLST visualisation using PHYLOViZ

- *S. pneumoniae* serotype 19A MLST genotypes
 - Grouped by lineage
 - Coloured by resistance profile
- PHYLOViZ visualisation



cgMLST visualisation using GrapeTree

- Click "Load Data"
- Select cgMLST95.tsv inside the cgmlst_matrix directory
- Select MSTreeV2 under the method drop down arrow

Tree Layout

Node Style

- Check Show Labels
- Colour By: ID
- Label Font Size: 11
- Node Size: 500
- Kurtosis (%): 30

Branch Style

- Check Show Labels
- Font Size: 17
- Scaling (%): 53
- Check Log Scale

Rendering

- Static

